Response to the Editor and Reviewers

We thank the editor and reviewers for the positive feedback and constructive suggestions. We present our responses to comments raised by each of the reviewers. Please note that we typeset the questions of reviewers in green while our answers are in black. Changes made to the original manuscript are highlighted in the updated version of our manuscript.

We hope that these responses satisfactorily address all of the comments raised, and look forward to participating in the conference.

Reviewer 1

The paper describes a model predictive control (MPC) scheme to optimize the operation of a battery system in the context of the balancing market. The authors model the market outcome within the MPC framework using an ICNN architecture, structured to enforce convexity and monotonicity with respect to the battery charging and discharging, resulting in a convex (relaxed) cost function. The approach is interesting and timely, and the paper is well written.
I have enjoyed reading the paper, interesting work.

Response: We would like to thank the reviewer for the encouraging and positive feedback.

Comment 1: The arguments of the function f_{market} in (9) and (10) do not agree with the original function definition in (7).

Response: We modified the inputs to the function f_{market} in eq. (7) based on the reviewer's suggestion, to make eq. (7) consistent with eqs. (9) and (10).

Comment 2: The title of the section V.A (experimental setup) should be revised because you are not proposing an experiment, that will be a simulation.

Response: We changed the title of Section V.A according to the reviewer's suggestion.

Comment 3: I assumed the notation y arrow is a vector. ϵ might be a vector too.

Response: As the reviewer pointed out correctly, the arrow notation indicates that the variable is a vector. To maintain consistency throughout Section II, we added arrow notation to ϵ as well

Comment 4: The authors initially treat the binary charging/discharging variables as input parameters in order to enforce convexity, but they ultimately solve a MIQP in which these binaries are decision variables. The logic of this narrative (i.e., first presenting the binaries as fixed parameters and later turning them into variables) is not clear to me. Moreover, there is no discussion of the value that convex cost functions and constraints provide in terms of tractability and problem structure (even when binary variables are present), and in the end the overall problem remains non-convex. Some rephrasing or restructuring in Section II could help clarify the authors' intent and the expectations regarding problem convexity.
On a similar note as above, the statement at the end of Section IV ("Because the optimization problem is convex [...]") is not correct. The problem, because it includes binary variables, is intrinsically non-convex. The fact that it can be then solved to optimality (and the structure will certainly help with that) is another matter (depending also on the optimality gap settings and whether the solution exists or not). Some more cautious rephrasing might help clarifying the attributes of the problem.

Response: We thank the reviewer for these highly pertinent comments, and updated the paper to clarify our approach. We agree with the reviewer that the main problem, defined by eqs. (2)–(6) and (8)–(10), remains non-convex. The primary motivation for convexifying the objective function is to guarantee that an optimal solution can be found. Although the overall problem is still non-convex, the number of binary variables is relatively small — i.e., equal to the look-ahead horizon length and the binaries obtained from the reformulation of the ICNN. Therefore, divide-and-conquer methods such as the branch-and-bound algorithm can guarantee convergence to the optimal solution. Moreover, the binary parameters in e qs. (9) and (10) are sort of flags (not decision variables) to calculate prices in two cases (charging/ discharging). We revised Section IV to better clarify the problem attributes and the expectations regarding problem convexity.

Comment 5: Regarding the battery model you use, it assumes several simplifying characteristics, such as constant efficiency across all power levels, zero reactive power, and unchanged power capability at any state of charge. The literature provides more comprehensive models, some of which also retain convexity, that might capture these effects more accurately.

Response: Adopting more detailed battery models can indeed be a powerful extension of the current work and an interesting direction for future research. We have now pointed this out at the end of the conclusion section, where future work is discussed.

Reviewer 2

The Authors present a data driven market model for implicit balancing in the form of input convex neural network. Because the output is convex with respect to the input, the implicit balancing problem can then be formulated as a convex MIQP with guaranteed global optimum. The complexity of the ICNN is kept in check by an internal attention mechanism that filters the bids via current and historic record of imbalances, so that only the most relevant bids are retained. The actions of the battery are determined via ICCN-assisted MPC and are demonstrated to outperform in terms of profit the MPC with baseline market clearing prices. The test results are produced in a broad variety of scenarios, that confirms the performance in presence of uncertainty in the imbalance forecast and for various battery sizes, and with the battery as price-taker and price-maker. Some consideration on the design parameters of the ICNN in relation to the resource type and the features of the historic data, both prices and imbalance, would be useful.

Response: We would like to thank the reviewer for the positive feedback.